

Project Name

QuickLoan – Scalable Web Application on AWS

Role: AWS Engineer

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ABSTRACT

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QuickLoan is a web-based loan application platform:

Backend: PHP running on Apache

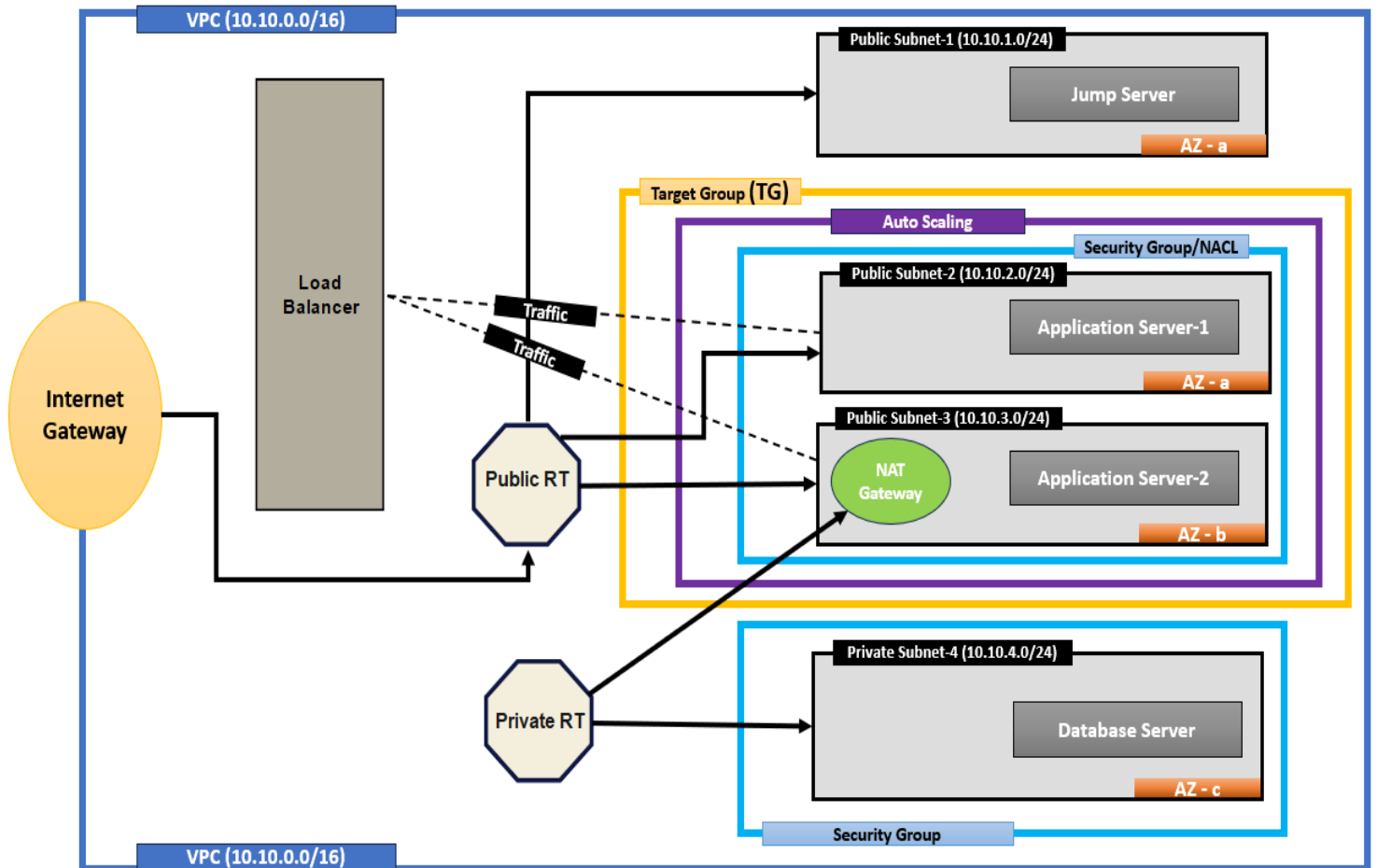
Frontend: HTML, CSS, JS

We architected and deployed QuickLoan on AWS with a fully functional VPC including public and private subnets. The backend EC2 instances securely communicated with an RDS MySQL database hosted in a private subnet with its own Security Group.

We deployed the app on an EC2 instance (Apache + PHP), validated it, and then created an AMI from that instance. This AMI was used in a Launch Template attached to an Auto Scaling Group to automatically scale instances based on demand (CPU usage, request count, etc.). An Application Load Balancer (ALB) was configured to distribute incoming requests across instances in multiple Availability Zones, ensuring high availability and fault tolerance.

SYSTEM ARCHITECTURE

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END-TO-END INFRASTRUCTURE BUILD PROCESS

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1. Create custom VPC

Your VPCs

[VPCs](#) | VPC encryption controls

Your VPCs (2) [Info](#)

Last updated less than a minute ago [Actions](#) [Create VPC](#)

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	vpc-1	vpc-0e0dec59c5bc2ed57	Available	Off	10.10.0.0/16	-

2. Create subnets- 3 public subnets and 1 private subnet.

Subnets (7) [Info](#)

Last updated 1 minute ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC	Block Pub...	IPv4 CIDR
☐	Public subnet-1	subnet-080e608ccdcd25c2	Available	vpc-0e0dec59c5bc2ed57 vpc-1	Off	10.10.1.0/24
☐	Public subnet-2	subnet-0a61ea838a4642dc5	Available	vpc-0e0dec59c5bc2ed57 vpc-1	Off	10.10.2.0/24
☐	Public subnet-3	subnet-0138a3e0a40485807	Available	vpc-0e0dec59c5bc2ed57 vpc-1	Off	10.10.3.0/24
☐	Private-subnet-4	subnet-0dc2d52e8a8e10794	Available	vpc-0e0dec59c5bc2ed57 vpc-1	Off	10.10.4.0/24

3. Create Internet gateway and attach to VPC

Internet gateway igw-0f1fa16206ad16589 successfully attached to vpc-0e0dec59c5bc2ed57

igw-0f1fa16206ad16589 / igw-1 [Actions](#)

Details [Info](#)

Internet gateway ID igw-0f1fa16206ad16589	State Attached	VPC ID vpc-0e0dec59c5bc2ed57 vpc-1	Owner 183425736765
--	-------------------	---	---------------------------------------

Tags (1) [Manage tags](#)

Search tags

Key	Value
Name	igw-1

4. Create NAT gateway and attach to public subnet-3 and attach Elastic IP to it.

NAT gateway nat-02b872c730a7795ae | NAT-gateway-1 was created successfully.

nat-02b872c730a7795ae / NAT-gateway-1

Actions

Details

NAT gateway ID nat-02b872c730a7795ae	Connectivity type Public	State Pending	State message Info -
NAT gateway ARN arn:aws:ec2:us-east-2:183425736765:natgateway/nat-02b872c730a7795ae	Primary public IPv4 address -	Primary private IPv4 address -	Primary network interface ID -
VPC vpc-0e0dec59c5bc2ed57 / vpc-1	Subnet subnet-0138a3e0a40485807 / Public subnet-3	Created Friday, 13 March 2026 at 11:34:39 GMT +5:30	Deleted -

Secondary IPv4 addresses Monitoring Attachments Tags

Secondary IPv4 addresses [Edit secondary IPv4 address associations](#)

Search

Private IPv4 address	Network interface ID	Status	Failure message
Secondary IPv4 addresses are not available for this nat gateway.			

5. Create public RT and private RT.

Route table rtb-0bd6ddb196a2e0be1 | Public-RT was created successfully.

rtb-0bd6ddb196a2e0be1 / Public-RT

Actions

Details [Info](#)

Route table ID rtb-0bd6ddb196a2e0be1	Main No	Explicit subnet associations -	Edge associations -
VPC vpc-0e0dec59c5bc2ed57 / vpc-1	Owner ID 183425736765		

Routes Subnet associations Edge associations Route propagation Tags

Routes (1) [Both](#) [Edit routes](#)

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.10.0.0/16	local	Active	No	Create Route Table

Route table rtb-05057e6fcc77e23e3 | Private-RT was created successfully.

rtb-05057e6fcc77e23e3 / Private-RT

Actions

Details [Info](#)

Route table ID rtb-05057e6fcc77e23e3	Main No	Explicit subnet associations -	Edge associations -
VPC vpc-0e0dec59c5bc2ed57 / vpc-1	Owner ID 183425736765		

Routes Subnet associations Edge associations Route propagation Tags

Routes (1) [Both](#) [Edit routes](#)

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.10.0.0/16	local	Active	No	Create Route Table

6. Attach public RT with all three public subnets.

✓ You have successfully updated subnet associations for rtb-0bd6ddb196a2e0be1 / Public-RT.

Route tables (1/4) Info Last updated less than a minute ago Actions Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associations	Edge associations	Main	VPC
☐	Default	rtb-096e764d4ec538468	-	-	Yes	vpc-06fe49cc
☐	-	rtb-05d87f70b3ef2adfc	-	-	Yes	vpc-0e0dec5f
☒	Public-RT	rtb-0bd6ddb196a2e0be1	3 subnets subnet-0a61ea838a4642dc5 / Public subnet-2 subnet-0138a3e0a40485807 / Public subnet-3 subnet-080e608ccdc25c2 / Public subnet-1	-	No	vpc-0e0dec5f
☐	Private-RT	rtb-05057e6fcc77e23e3	-	-	No	vpc-0e0dec5f

7. Attach private RT with private subnet.

✓ You have successfully updated subnet associations for rtb-05057e6fcc77e23e3 / Private-RT.

Route tables (1/4) Info Last updated less than a minute ago Actions Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associations	Edge associations	Main	VPC
☐	Default	rtb-096e764d4ec538468	-	-	Yes	vpc-06fe49cc
☐	-	rtb-05d87f70b3ef2adfc	-	-	Yes	vpc-0e0dec5f
☐	Public-RT	rtb-0bd6ddb196a2e0be1	3 subnets	-	No	vpc-0e0dec5f
☒	Private-RT	rtb-05057e6fcc77e23e3	subnet-0dc2d52e8a8e10794 / Private-subnet-4	-	No	vpc-0e0dec5f

8. Attach public RT with Internet Gateway (IGW)

✓ Updated routes for rtb-0bd6ddb196a2e0be1 / Public-RT successfully
Details

rtb-0bd6ddb196a2e0be1 / Public-RT Actions

Details Info

Route table ID rtb-0bd6ddb196a2e0be1	Main No	Explicit subnet associations 3 subnets	Edge associations -
VPC vpc-0e0dec59c5bc2ed57 vpc-1	Owner ID 183425736765		

Routes Subnet associations Edge associations Route propagation Tags

Routes (2) Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0f1fa16206ad16589	Active	No	Create Route
10.10.0.0/16	local	Active	No	Create Route Table

9. Attach private RT with NAT Gateway

Updated routes for rtb-05057e6fcc77e23e3 / Private-RT successfully
Details

rtb-05057e6fcc77e23e3 / Private-RTActions

DetailsInfo

Route table ID
rtb-05057e6fcc77e23e3

VPC
vpc-0e0dec59c5bc2ed57 | vpc-1

Main
No

Owner ID
183425736765

Explicit subnet associations
subnet-0dc2d52e8a8e10794 / Private-subnet-4

Edge associations
-

RoutesSubnet associationsEdge associationsRoute propagationTags

Routes (2)

Filter routes

BothEdit routes

< 1 > ⚙

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-02b872c730a7795ae	Active	No	Create Route
10.10.0.0/16	local	Active	No	Create Route Table

10. Create two Security Group (SG groups):

(a) 1 Security Group for application servers (SG-Application-servers).

Security group (sg-05c17a55a65519e1c | SG-Application-server) was created successfully
Details

sg-05c17a55a65519e1c - SG-Application-serverActions

Details

Security group name
SG-Application-server

Owner
183425736765

Security group ID
sg-05c17a55a65519e1c

Inbound rules count
2 Permission entries

Description
SG-Application-server

Outbound rules count
1 Permission entry

VPC ID
vpc-0e0dec59c5bc2ed57

Inbound rulesOutbound rulesSharingVPC associationsRelated resources - newTags

Inbound rules (2)

Search

Manage tagsEdit inbound rules

< 1 > ⚙

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	S
☐	-	sgr-038c7012af0aec59d	IPv4	HTTP	TCP	80	C
☐	-	sgr-00b5936af50f713f5	IPv4	SSH	TCP	22	C

(b) 1 Security Group for database servers (SG-Database-server).

Security group (sg-0d914bcdd950358c4 | SG-Database-server) was created successfully

Details

sg-0d914bcdd950358c4 - SG-Database-server

Actions

Details

Security group name
SG-Database-server

Security group ID
sg-0d914bcdd950358c4

Description
SG-Database-server

VPC ID
vpc-0e0dec59c5bc2ed57

Owner
183425736765

Inbound rules count
2 Permission entries

Outbound rules count
1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (2)

Manage tags

Edit inbound rules

Search

	Name	Security group rule ID	IP version	Type	Protocol	Port range
☐	-	sgr-0e1914ea9f5c4684c	IPv4	SSH	TCP	22
☐	-	sgr-0e627c6fde0620fe8	IPv4	MYSQL/Aurora	TCP	3306

11.Create 1 Jump server in public subnet-1 (AZ-a) and attach SG

Instances (1/1) Info

Last updated 1 minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☒	Jump-server	i-093e33f5f88c7761f	Running	t3.micro	Initializing	View alarms +	us-east-2a	-

i-093e33f5f88c7761f (Jump-server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

VPC ID
vpc-0e0dec59c5bc2ed57 (vpc-1)

Subnet ID
subnet-080e608ccddced25c2 (Public subnet-1)

Availability zone
us-east-2a

Availability zone ID
use2-az1

Outpost ID
-

IP addresses

Public IPv4 address
3.145.93.46 | open address

Private IPv4 addresses
10.10.1.128

IPv6 addresses
-

Secondary private IPv4 addresses
-

Carrier IP addresses (ephemeral)
-

Hostname and DNS

Public DNS
-

Private IP DNS name (IPv4 only)
ip-10-10-1-128.us-east-2.compute.internal

IPv4-only IP based name: A record only
-

i-093e33f5f88c7761f (Jump-server)

183425736765

Fri Mar 13 2026 12:03:48 GMT+0530 (India Standard Time)

Security groups

sg-05c17a55a65519e1c (SG-Application-server)

Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Descr
-	sgr-038c7012af0aec59d	80	TCP	0.0.0.0/0	SG-Application-server	-
-	sgr-00b5936af50f713f5	22	TCP	0.0.0.0/0	SG-Application-server	-

12. Create application server each in public subnet-2 (AZ-a) & public subnet-3 (AZ-b)

(a) Application server-1 in public subnet-2 (AZ-a)

The screenshot displays the AWS Management Console interface for the 'Instances' page. At the top, there's a search bar and a table of instances. The table has columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4 DNS. The instance 'Application-Server-1' with ID 'i-0511a458c0715d07d' is highlighted. Below the table, the details for 'i-0511a458c0715d07d (Application-Server-1)' are shown under the 'Networking' tab. The details include VPC ID (vpc-0e0dec59c5bc2ed57), Subnet ID (subnet-0a61ea838a4642dc5), Availability zone (us-east-2a), Public IPv4 address (16.59.125.3), Private IPv4 addresses (10.10.2.75), and Private IP DNS name (ip-10-10-2-75.us-east-2.compute.internal).

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
Jump-server	i-093e33f5f88c7761f	Running	t3.micro	3/3 checks passed	View alarms +	us-east-2a	-
Application-Server-1	i-0511a458c0715d07d	Running	t3.micro	Initializing	View alarms +	us-east-2a	-

i-0511a458c0715d07d (Application-Server-1)

Details | Status and alarms | Monitoring | Security | **Networking** | Storage | Tags

VPC ID: vpc-0e0dec59c5bc2ed57 (vpc-1)

Subnet ID: subnet-0a61ea838a4642dc5 (Public subnet-2)

Availability zone: us-east-2a

Outpost ID: -

Public IPv4 address: 16.59.125.3 | open address

Private IPv4 addresses: 10.10.2.75

IPv6 addresses: -

Secondary private IPv4 addresses: -

Carrier IP addresses (ephemeral): -

Public DNS: -

Private IP DNS name (IPv4 only): ip-10-10-2-75.us-east-2.compute.internal

IPv4-only IP based name: A record only

(b) Application server-2 in public subnet-3 (AZ-b)

The screenshot displays the AWS Management Console interface for the 'Instances' page. The table of instances shows 'Application-Server-2' with ID 'i-084cd38e6b42887c6' in a 'Running' state. Below the table, the details for 'i-084cd38e6b42887c6 (Application-Server-2)' are shown under the 'Networking' tab. The details include VPC ID (vpc-0e0dec59c5bc2ed57), Subnet ID (subnet-0138a3e0a40485807), Availability zone (us-east-2b), Public IPv4 address (3.134.245.106), Private IPv4 addresses (10.10.3.100), and Private IP DNS name (ip-10-10-3-100.us-east-2.compute.internal).

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
Application-Server-2	i-084cd38e6b42887c6	Running	t3.micro	Initializing	View alarms +	us-east-2b	-
Jump-server	i-093e33f5f88c7761f	Stopped	t3.micro	-	View alarms +	us-east-2a	-
Application-Server-1	i-0511a458c0715d07d	Stopped	t3.micro	-	View alarms +	us-east-2a	-

i-084cd38e6b42887c6 (Application-Server-2)

Details | Status and alarms | Monitoring | Security | **Networking** | Storage | Tags

VPC ID: vpc-0e0dec59c5bc2ed57 (vpc-1)

Subnet ID: subnet-0138a3e0a40485807 (Public subnet-3)

Availability zone: us-east-2b

Outpost ID: -

Public IPv4 address: 3.134.245.106 | open address

Private IPv4 addresses: 10.10.3.100

IPv6 addresses: -

Secondary private IPv4 addresses: -

Carrier IP addresses (ephemeral): -

Public DNS: -

Private IP DNS name (IPv4 only): ip-10-10-3-100.us-east-2.compute.internal

IPv4-only IP based name: A record only

13. Create 1 database server in private subnet-4 (AZ-c)

The screenshot displays the AWS Management Console interface for an EC2 instance. At the top, the 'Instances (1/4)' page is shown with a search bar and filters. The instance 'Database-server' (ID: i-0c95ee6d35a2b03e4) is listed in a 'Running' state, using a 't3.micro' instance type in the 'us-east-2c' availability zone. Below the instance list, the 'Networking' tab is selected for the instance details. The details show the VPC ID as 'vpc-0e0dec59c5bc2ed57', Subnet ID as 'subnet-0dc2d52e8a8e10794' (Private-subnet-4), and Availability zone as 'us-east-2c'. The IP addresses section shows a Private IPv4 address of '10.10.4.251' and a Private IP DNS name of 'ip-10-10-4-251.us-east-2.compute.internal'.

Instances (1/4)							
Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
Database-server	i-0c95ee6d35a2b03e4	Running	t3.micro	Initializing	View alarms +	us-east-2c	-

i-0c95ee6d35a2b03e4 (Database-server)		
Networking		
VPC ID vpc-0e0dec59c5bc2ed57 (vpc-1)	Subnet ID subnet-0dc2d52e8a8e10794 (Private-subnet-4)	Availability zone us-east-2c
Availability zone ID use2-az3	Outpost ID -	
Public IPv4 address -	Private IPv4 addresses 10.10.4.251	IPv6 addresses -
Secondary private IPv4 addresses -	Carrier IP addresses (ephemeral) -	
Public DNS -	Private IP DNS name (IPv4 only) ip-10-10-4-251.us-east-2.compute.internal	IPv4-only IP based name: A record only -

14. Create 1 database in RDS

The screenshot shows the AWS Management Console interface for creating a new RDS database instance. A blue banner at the top indicates 'Creating database customersdb' and provides instructions on using settings from 'customersdb' to simplify configuration. Below the banner, the 'Databases (1)' page is shown with a search bar and filters. The instance 'customersdb' is listed in a 'Creating' state, using a 'MySQL Co...' engine, 'SECOND' upgrade rollout order, in the 'us-east-2c' region, with a size of 'db.t4g.micro'.

Databases (1)							
DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size	Recommendation
customersdb	Creating	Instance	MySQL Co...	SECOND	us-east-2c	db.t4g.micro	

15. Connect Jump server through putty

```
root@ip-10-10-1-128:~  
login as: ec2-user  
Authenticating with public key "Jump-server-key"  
  
#  
~\###_ Amazon Linux 2023  
~~\#####\  
~~\_####|  
~~\_#/ https://aws.amazon.com/linux/amazon-linux-2023  
~~V~'-'>  
~~~~  
~~.-.  
~/m/'
```

[ec2-user@ip-10-10-1-128 ~]\$ sudo su -
[root@ip-10-10-1-128 ~]#

16. Create key pair files in Jump server

```
root@ip-10-10-1-128:~  
[root@ip-10-10-1-128 ~]# ls  
Application-server-key.pem  Database-server-key.pem  
[root@ip-10-10-1-128 ~]#
```

17. Grant permission to keypair files: CMD – `chmod 400 <keypair name>`

```
root@ip-10-10-1-128:~  
[root@ip-10-10-1-128 ~]# chmod 400 Database-server-key.pem  
[root@ip-10-10-1-128 ~]# chmod 400 Application-server-key.pem  
[root@ip-10-10-1-128 ~]#
```

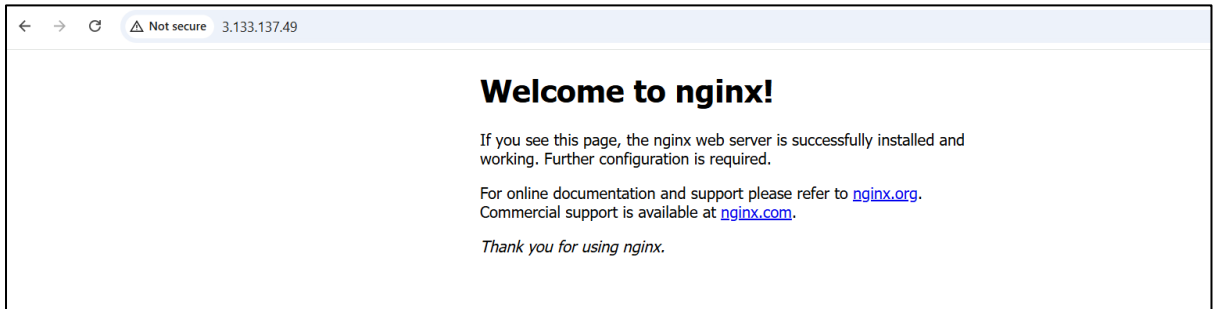
18. Jump into application server from Jump server

```
[root@ip-10-10-2-75:~  
[root@ip-10-10-1-128 ~]# chmod 400 Database-server-key.pem  
[root@ip-10-10-1-128 ~]# chmod 400 Application-server-key.pem  
[root@ip-10-10-1-128 ~]# ssh -i Application-server-key.pem ec2-user@10.10.2.75  
The authenticity of host '10.10.2.75 (10.10.2.75)' can't be established.  
ED25519 key fingerprint is SHA256:pXT3sRhRh6Xn1LtPuV7LxfsmLM4+zMTTricSDfbwUC5Q.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.10.2.75' (ED25519) to the list of known hosts.  
  
      #_ _  
    ~\   #####          Amazon Linux 2023  
~~~ \___#####\  
~~~~ \____|  
~~~~ \#/           https://aws.amazon.com/linux/amazon-linux-2023  
~~~~ V~'|'->  
~~~~  
~~~~_.._/_/_/_/  
_____/_m/'/_/_/_/  
[ec2-user@ip-10-10-2-75 ~]$ sudo su -  
[root@ip-10-10-2-75 ~]#
```

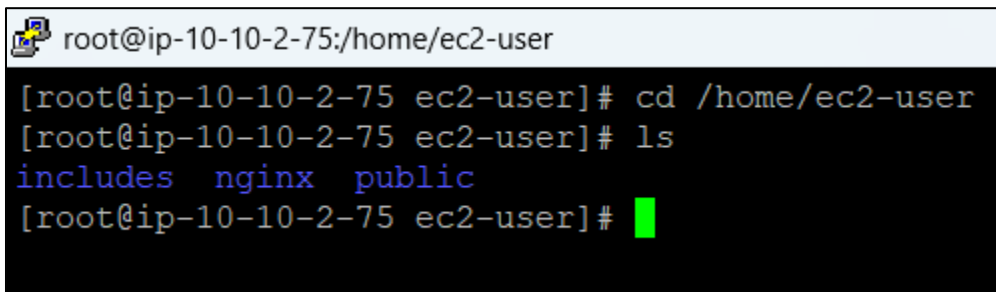
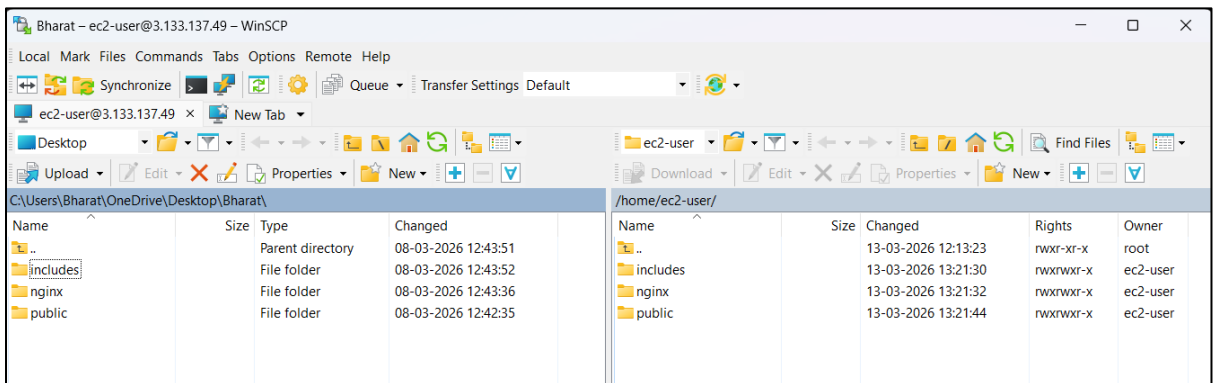
19. Install all dependent package for “nginx” & “php”

```
root@ip-10-10-2-75:~  
[root@ip-10-10-2-75 ~]# systemctl restart nginx  
[root@ip-10-10-2-75 ~]# history  
  1  yum update -y  
  2  yum install -y nginx  
  3  systemctl start nginx  
  4  chkconfig nginx on  
  5  yum install -y php8.2 php-fpm php-mysqlnd php-pdo php-mbstring  
  6  php -v  
  7  systemctl start php-fpm  
  8  chkconfig php-fpm on  
  9  systemctl restart nginx  
 10  history  
[root@ip-10-10-2-75 ~]#
```

20. Cross check whether nginx is working or not by entering publicip:80 in the browser



21. Copy the following three folders from our laptop (windows) to the application server (Linux) using WinSCP. (includes, nginx, public)



22. Move Includes & Public folders to HTML folder by running below CMD:

```
mv /home/ec2-user/includes /usr/share/nginx/html
```

```
mv /home/ec2-user/public /usr/share/nginx/html
```

```
root@ip-10-10-2-75:/usr/share/nginx/html
[root@ip-10-10-2-75 ec2-user]# cd /usr/share/nginx/html
[root@ip-10-10-2-75 html]# ls
404.html  50x.html  icons  includes  index.html  nginx-logo.png  poweredby.png  public
[root@ip-10-10-2-75 html]#
```

23. Update owner name and group name of public folder from ec2-user to nginx. Run CMD:

```
chown -R nginx:nginx /usr/share/nginx/html/public
```

From

```
drwxrwxr-x. 3 ec2-user ec2-user 103 Mar 13 07:51 public
[root@ip-10-10-2-75 html]#
```

To

```
drwxrwxr-x. 3 nginx nginx 103 Mar 13 07:51 public
[root@ip-10-10-2-75 html]#
```

24. Grant permission to public folder by running below CMD:

```
chmod -R 755 /usr/share/nginx/html/public
```

```
root@ip-10-10-2-75:/usr/share/nginx/html
[root@ip-10-10-2-75 html]# chmod -R 755 /usr/share/nginx/html/public
[root@ip-10-10-2-75 html]#
```

25. Create 1 bucket in S3 (aws-project-bucket-13032026)

Amazon S3 > Buckets

✔ Successfully created bucket "aws-project-bucket-13032026"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

Buckets

[General purpose buckets](#) [All AWS Regions](#) [Directory buckets](#)

General purpose buckets (1) [Info](#) [Refresh](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
aws-project-bucket-13032026	US East (Ohio) us-east-2	March 13, 2026, 19:01:49 (UTC+05:30)

26. Upload "images" folder in S3 bucket which is available in "public" folder.

Amazon S3

✔ Upload succeeded
For more information, see the [Files and folders](#) table.
After you navigate away from this page, the following information is no longer available.

Summary

Destination: [s3://aws-project-bucket-13032026](#)

Succeeded: 5 files, 527.1 KB (100.00%)

Failed: 0 files, 0 B (0%)

[Files and folders](#) [Configuration](#)

Files and folders (5 total, 527.1 KB)

Find by name

Name	Folder	Type	Size	Status	Error
gold_loan.jpg	images/	image/jpeg	142.6 KB	✔ Succeeded	-
home_loan.jpg	images/	image/jpeg	26.7 KB	✔ Succeeded	-
personal_loan.jpg	images/	image/jpeg	117.0 KB	✔ Succeeded	-
quickloan_logo.png	images/	image/png	16.7 KB	✔ Succeeded	-
vehicle_loan.jpg	images/	image/jpeg	224.1 KB	✔ Succeeded	-

27. Update the image paths from the S3 bucket in the index.html file.

(a) First index.html file using path: vim

/usr/share/nginx/html/public/index.html

(b) And copy images source URL (old):

"https://loanapp332211.s3.eu-west-2.amazonaws.com/images/quickloan_logo.png"

(c) And copy any random images URL from bucket (latest):

https://aws-project-bucket-13032026.s3.us-east-2.amazonaws.com/images/gold_loan.jpg

(d) Now search old URL and replace with latest URL in html file code, using below CMD:

:%s/loanapp332211.s3.eu-west-2.amazonaws.com/aws-project-bucket-13032026.s3.us-east-2.amazonaws.com

(This will make changes in all 5 available image source URL in index.html file code)

(e) Copy CMD and paste it in index.html file. Save the file and exit.

:%s/loanapp332211.s3.eu-west-2.amazonaws.com/aws-project-bucket-13032026.s3.us-east-2.amazonaws.com

With old URL

```
root@ip-10-10-2-75:~
!DOCTYPE html
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>QuickLoan - Your Trusted Financial Partner</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <div class="logo">
      
    </div>
    <nav>
      <a href="apply.php" class="apply-btn">Apply for Loan</a>
    </nav>
  </header>
  <section class="hero">
    <h1>Your Financial Dreams, Realized.</h1>
    <p>QuickLoan provides fast, reliable, and affordable financial solutions.</p>
  </section>
  <section class="products">
```

With latest URL

```
root@ip-10-10-2-75:/usr/share/nginx/html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>QuickLoan - Your Trusted Financial Partner</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <div class="logo">
      
    </div>
  </header>
  <nav>
```

By these steps we have defined/changed location of images files from old path to the latest location path where image files are uploaded in S3 bucket.

28. Update quick loan image location in “apply.php” file which is available under “Public folder”

- (a) Open file using path CMD: `vim /usr/share/nginx/html/public/apply.php`
- (b) Need to update file location (old) shown in below screenshot:

images/quickloan_logo.png

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Apply for Loan - QuickLoan</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <div class="logo">
      
    </div>
  </header>
  <nav>
```

- (c) Copy URL of “quickloan_logo” image:
`https://aws-project-bucket-13032026.s3.us-east-2.amazonaws.com/images/quickloan_logo.png`

(d) Edit file by pressing 'I' button, remove highlighted old URL and paste new copied URL and save the file.

See the changes:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Apply for Loan - QuickLoan</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <div class="logo">
      
    </div>
  </header>
  <section class="apply-form">
```

29. Navigate to "Includes" folder and open "db_connect.php" file. Will update credentials to connect webserver with database.

(a) Run CMD: `cd /usr/share/nginx/html/includes`

(b) Can see file named: **vim db_connect.php**

```
[root@ip-10-10-2-75 html]# cd /usr/share/nginx/html/includes
[root@ip-10-10-2-75 includes]# ls
db_connect.php
[root@ip-10-10-2-75 includes]#
```

(c) Open file using CMD: `vim db_connect.php`

And update below given credentials:

- Servername: enter here database endpoint detail
- Username: let it be as it is, as we have set user name as "admin"
- Password: let it be as it is, as we have set password as "admin123"
- dbname: let it be as it is, as we have set database name as "quickloan_db"

Once changes done save file.

```
?php
$servername = "loanform1.c74u6oq6k99z.eu-west-2.rds.amazonaws.com";
$username = "admin";
$password = "admin123";
$dbname = "quickloan_db";
```

30.Navigate to Jump server from application server by entering exit 2 times.

```
[root@ip-10-10-2-75 includes]# exit
logout
[ec2-user@ip-10-10-2-75 ~]$ exit
logout
Connection to 10.10.2.75 closed.
[root@ip-10-10-1-128 ~]#
```

31.Navigate to Database server from Jump server by running below CMD:

```
ssh -i <db server keypair name> ec2-user@pvtip
```

```
[root@ip-10-10-1-128 ~]# ssh -i Database-server-key.pem ec2-user@10.10.4.251
The authenticity of host '10.10.4.251 (10.10.4.251)' can't be established.
ED25519 key fingerprint is SHA256:49FGAqEm9wFuSzOW3ZygGy70LFrvSnnXjvORukX2XIs.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.4.251' (ED25519) to the list of known hosts.

      #
    ~\  ####_      Amazon Linux 2023
  ~ ~ \  #####\
  ~ ~  \####|
  ~ ~   \#/      https://aws.amazon.com/linux/amazon-linux-2023
    ~ ~   V~'  ->
      ~ ~ ~
      ~ ~ . _ . /
      ~ ~ _ /m/ ' ->
[ec2-user@ip-10-10-4-251 ~]$ sudo su -
[root@ip-10-10-4-251 ~]#
```

32.Install database package: Mariadb105

CMD - yum install mariadb105 -y

33. Enter into database application (MySQL)

CMD - `mysql -h endpoint -u username -ppassword`

```
[root@ip-10-10-4-251 ~]# mysql -h customersdb.ct2ocgamcrah.us-east-2.rds.amazonaws.com -u admin -ppadmin123
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 287
Server version: 8.4.7 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>
```

34. Enter into database which was created in advance: quickloan_db

Query – use quickloan_db;

```
MySQL [(none)]> use quickloan_db;
Database changed
MySQL [quickloan_db]>
```

35. Create a table name as “applications”

(Make sure create table as per schema defined in index.html code)

Query as below given –

```
CREATE TABLE applications (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(255) NOT NULL,
  contact VARCHAR(20) NOT NULL,
  email VARCHAR(255) NOT NULL,
  loan_type VARCHAR(50) NOT NULL
);
```

```
MySQL [quickloan_db]> CREATE TABLE applications (
->   id INT AUTO_INCREMENT PRIMARY KEY,
->   name VARCHAR(255) NOT NULL,
->   contact VARCHAR(20) NOT NULL,
->   email VARCHAR(255) NOT NULL,
->   loan_type VARCHAR(50) NOT NULL
-> );
Query OK, 0 rows affected (0.065 sec)

MySQL [quickloan_db]> show tables;
+-----+
| Tables_in_quickloan_db |
+-----+
| applications           |
+-----+
1 row in set (0.001 sec)
```

36. Navigate to Jump server from database server by entering exit 3 times.

- (a) Exit; – database application (MySQL)
- (b) Exit – left root user and entered in ec2-user
- (c) Exit - database server and entered in Jump server (shows Jump server PVT IP)

```
MySQL [quickloan_db]> exit
Bye
[root@ip-10-10-4-251 ~]# exit
logout
[ec2-user@ip-10-10-4-251 ~]$ exit
logout
Connection to 10.10.4.251 closed.
[root@ip-10-10-1-128 ~]#
```


SAMPLE CODING

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PHP Code for Website

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Apply for Loan - QuickLoan</title>

  <link rel="stylesheet" href="styles.css">

</head>

<body>

  <header>

    <div class="logo">

    </div>

  </header>

  <section class="apply-form">

    <h2>Apply for Loan</h2>

    <form action="submit_application.php" method="post">

      <label for="name">Name:</label>

      <input type="text" id="name" name="name" required>

      <label for="contact">Contact Number:</label>

      <input type="tel" id="contact" name="contact" required>

    </form>

  </section>

</body>

</html>
```

```
<label for="email">Email Address:</label>

<input type="email" id="email" name="email" required>


<label for="loan_type">Type of Loan:</label>

<select id="loan_type" name="loan_type" required>

  <option value="Home Loan">Home Loan</option>

  <option value="Gold Loan">Gold Loan</option>

  <option value="Vehicle Loan">Vehicle Loan</option>

  <option value="Personal Loan">Personal Loan</option>

</select>


<div class="form-buttons">

  <button type="submit" class="submit-btn">Submit</button>

  <button type="reset" class="reset-btn">Reset</button>

</div>

</form>

</section>

<script>

  document.querySelector('form').addEventListener('submit', function(event) {

    alert('Application submitted successfully. Our representative will contact you shortly.');
```

});

```
</script>

</body>

</html>
```

TRAINING CERTIFICATE

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